

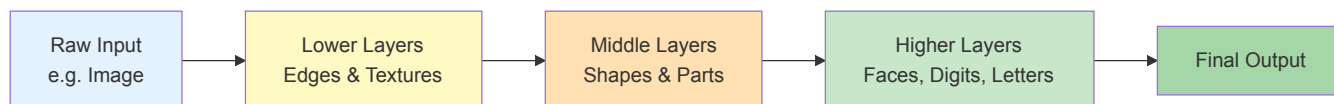
0. What is Deep Learning

What is Deep Learning?

Deep Learning is a subfield of Artificial Intelligence and Machine Learning, inspired by the structure of the human brain.

Deep Learning algorithms attempt to draw similar conclusions as humans would by continually analysing data through a logical structure called a **Neural Network**. It is part of a broader family of machine learning methods based on artificial neural networks with **representation learning**.

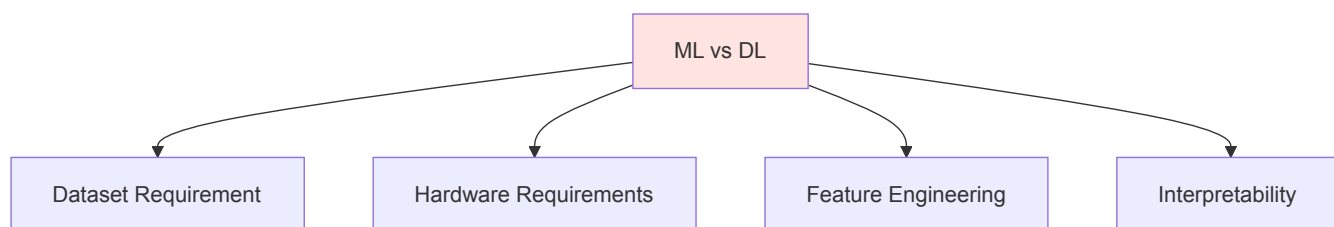
Deep Learning algorithms use multiple layers to progressively extract higher-level features from raw input.



How Does Deep Learning Differ from Machine Learning?

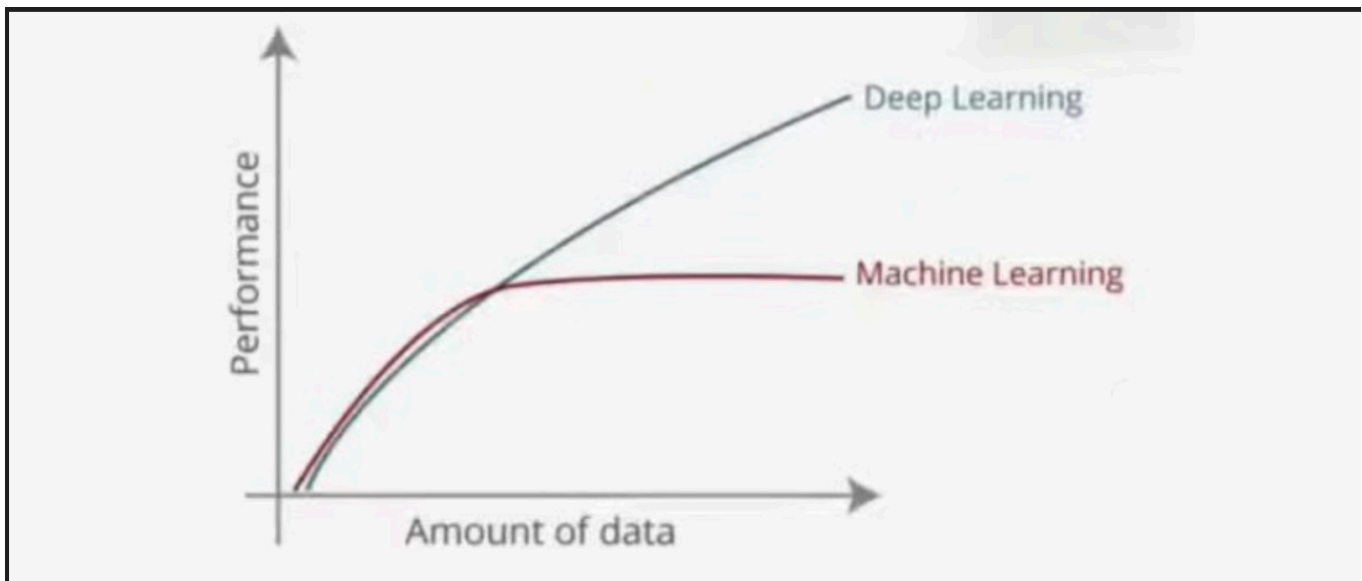
In Machine Learning, the relationship between input and output is decided through statistical analysis. In Deep Learning, this relationship is learned through a neural network.

Beyond that fundamental difference, the two diverge across several key dimensions:



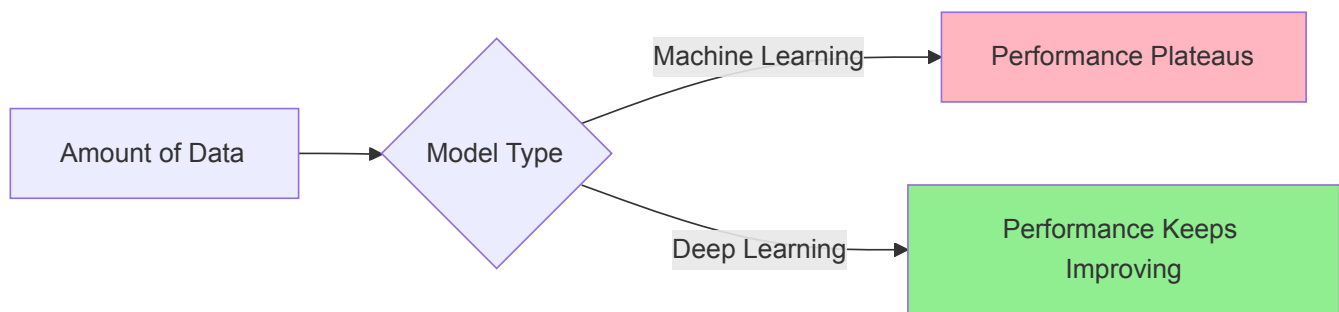
1. Dataset Requirement

Deep Learning models require significantly more data than Machine Learning models to perform well.



As the graph shows, after a certain volume of data, the performance of a Machine Learning model **stagnates** - it cannot extract more signal no matter how much more data you add. Deep Learning models, on the other hand, **continue to improve** as data increases.

This is why Deep Learning is described as being **data hungry**.

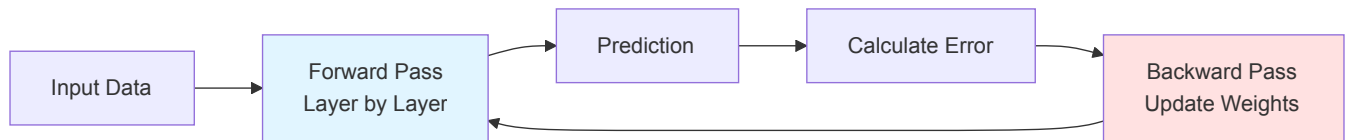


2. Hardware Requirements

	Machine Learning	Deep Learning
Hardware	CPU is sufficient	GPU required
Why	Statistical computations are sequential	Neural networks involve complex matrix multiplications that must run in parallel
Training Time	Minutes to hours	Hours to days to weeks

Deep Learning's computational core - forward and backward passes through layers of a neural network - is essentially massive matrix multiplication. GPUs are architecturally designed to

perform thousands of these operations simultaneously, which is why they are non-negotiable for DL training.

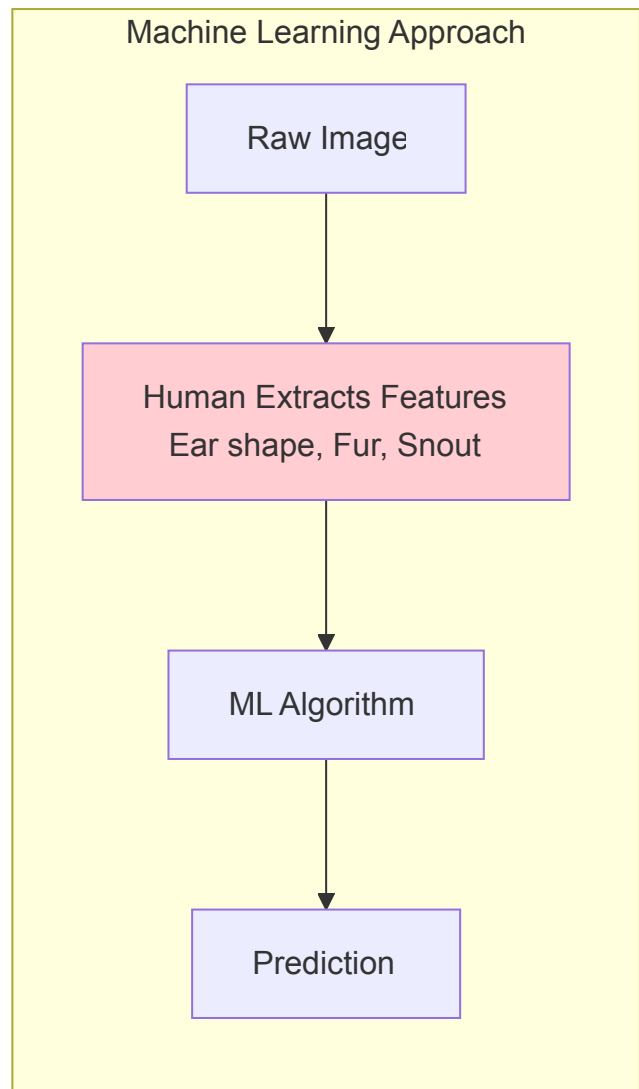
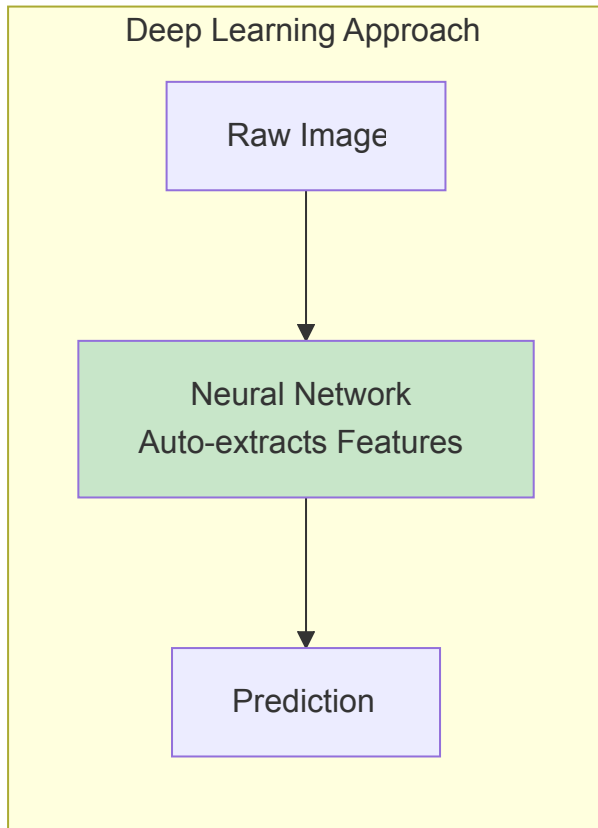


3. Feature Engineering

	Machine Learning	Deep Learning
Approach	Features must be manually engineered by a human	Features are automatically extracted from raw data
What this means	You decide what the model sees	The model decides what matters
Term for DL's approach	-	Representation Learning

Example - Dog vs Cat Classification:

- In **Machine Learning**: you must manually engineer features like ear shape, snout length, fur texture, etc. and feed those to the model
- In **Deep Learning**: you feed in the raw image and the network automatically learns to identify the relevant distinguishing features on its own



4. Interpretability / Explainability

Deep Learning models are often called a **black box**.

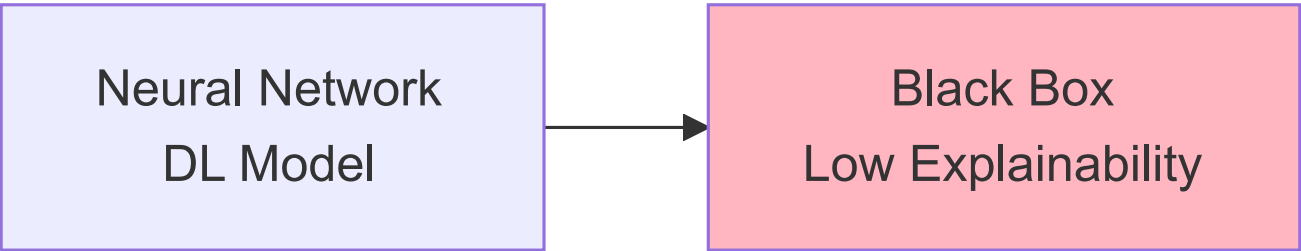
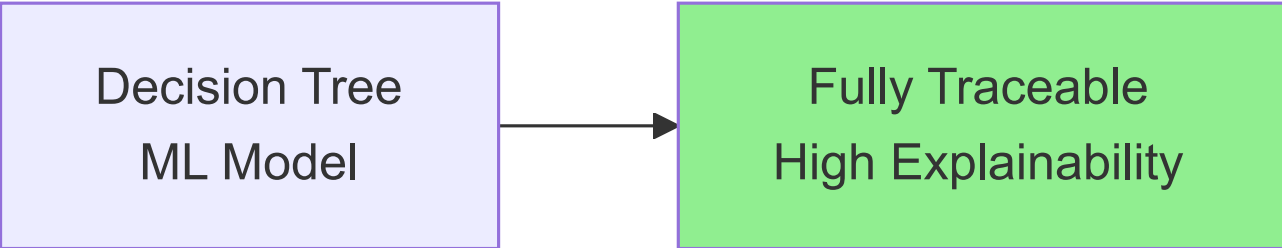
The model produces an output, but we have no clear understanding of *why* it produced that specific output or what internal reasoning led to it. If a DL model classifies an image as a dog, we cannot easily inspect what pattern in the image triggered that decision.

Machine Learning models, by contrast, tend to have **high explainability**. A Decision Tree, for example, can be visualised step by step - you can trace exactly which feature at which threshold led to any given prediction.

⚠ The Explainability Trade-off

This is one of the central tensions in modern AI. Deep Learning models are often the highest-performing option, but their black-box nature makes them difficult to audit, debug,

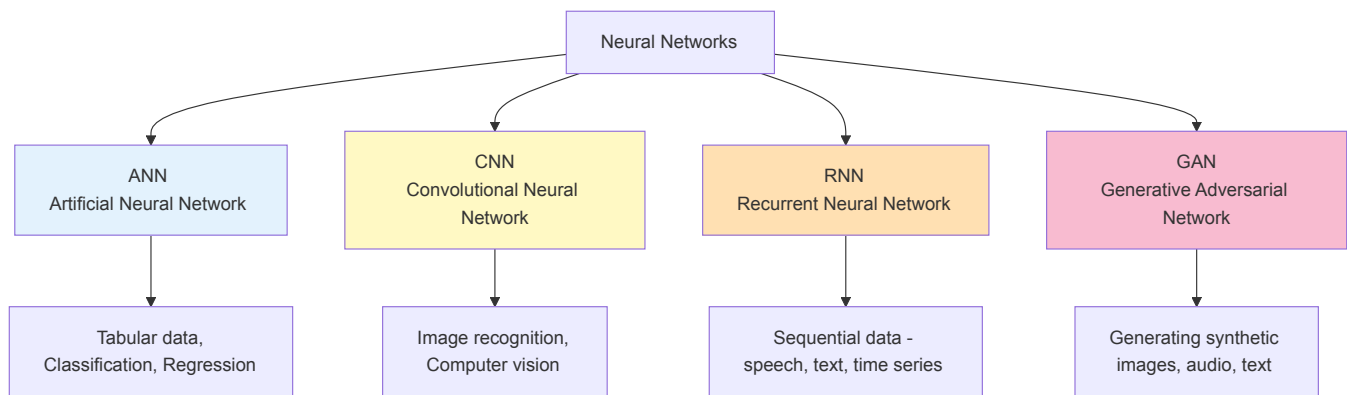
and deploy in regulated industries like healthcare, finance, and law where decisions must be justifiable.



Neural Networks

Neural networks are logical structures designed to replicate the way human brains process information - connecting inputs to outputs through layers of interconnected nodes.

There are several types of neural networks, each suited to different kinds of data and problems:

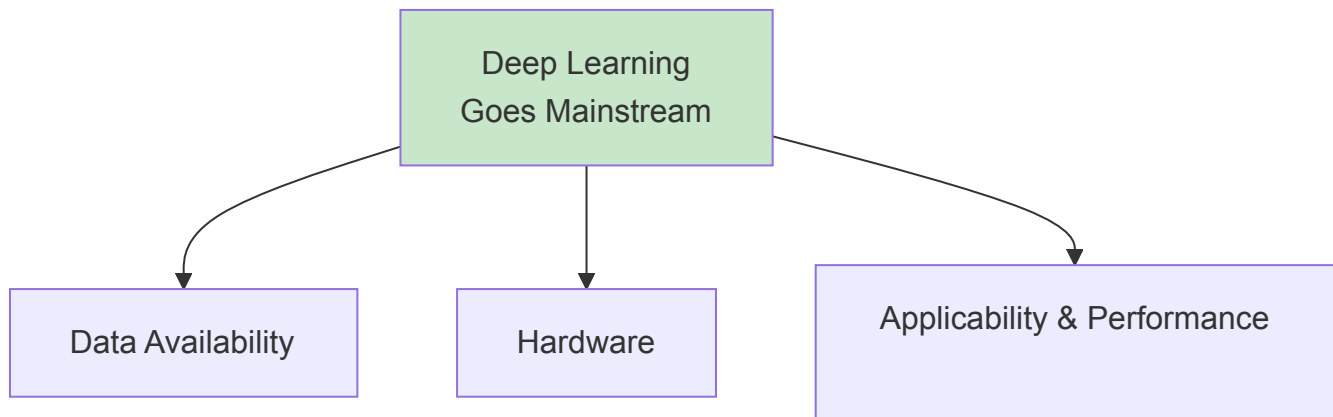


Network Type	Full Name	Primary Use Case
ANN	Artificial Neural Network	General-purpose - tabular data, classification, regression

Network Type	Full Name	Primary Use Case
CNN	Convolutional Neural Network	Image recognition, computer vision, image generation
RNN	Recurrent Neural Network	Sequential data - speech, text, time series
GAN	Generative Adversarial Network	Generating synthetic images, audio, and text

Why is Deep Learning Becoming Mainstream Now?

Deep Learning is not new - the foundational ideas have existed since the 1980s. What changed is the confluence of three things arriving simultaneously:



1. Data Availability

The internet generated an explosion of labelled data - images, text, audio, video - at a scale that simply did not exist before. Deep Learning's data hunger is finally being met.

2. Hardware

Modern GPUs (and more recently TPUs) made it economically feasible to train large neural networks in reasonable time. What once took weeks on a supercomputer now takes hours on a cloud GPU.

3. Applicability and Performance

Deep Learning began outperforming every prior approach on benchmark after benchmark - image classification (ImageNet 2012), speech recognition, machine translation, game playing. Once performance results became undeniable, industry adoption accelerated rapidly.

ML vs DL - Full Comparison

Dimension	Machine Learning	Deep Learning
Data requirement	Works with small to medium datasets	Requires large datasets
Hardware	CPU sufficient	GPU required
Training time	Minutes to hours	Hours to weeks
Feature engineering	Manual - human designed	Automatic - representation learning
Explainability	High - interpretable models exist	Low - black box
Performance ceiling	Plateaus with more data	Continues to improve with more data
Best for	Structured / tabular data	Images, text, audio, video